

Predicting Value with Accounting Data

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Summary

I tested whether cheap stocks, measured by the Price-to-Book (P/B) ratio, beat the market once I only keep the ones that actually make money. I backtested this on US stocks from March 2021 to September 2025 (18 quarters) in two ways: sector by sector against each sector's ETF, and as one portfolio from the 500 biggest US stocks against SPY.

Cheapness alone did not beat the market. Sector by sector, the cheapest-P/B baskets beat their sector ETF only 35% of the time, and the biggest wins came from one-off spikes in tiny penny stocks. After I removed the penny stocks and required a profit, the win rate fell to 13%. Some parameter settings did outperform their sector index (such as Basic Materials and Energy), but with only 18 quarterly samples the confidence in any single setting was low. Against SPY (13.81% a year), filters that only weed out bad stocks cut volatility and max drawdown below SPY, but never beat it on return. Only my strictest filters did: keeping only the top 25% most profitable companies (by ROA and ROE), and my strictest initial filters. But those left only about 7 stocks and about 2 stocks on average.

In my tests, how profitable a company was mattered more than how cheap it was. In my grid search (trying every combination of settings), the profit filter moved returns by about 8 points a year, while the P/B filter moved them by about 2.5. **The biggest lesson was about data.** While building the project, I ran into two kinds of data problems: real market extremes (like penny-stock spikes and "zombie" companies with negative equity) and corrupted vendor numbers (like share counts reported 1,000 times too high). Finding and fixing these taught me more than any single result.

Introduction

I want to see if raw accounting data can reliably show if a company is under or overvalued. This helps cut through daily market noise. I do not want to guess market sentiment or investor psychology. Instead, I use simple code. It checks if financial statement signals can systematically find strong but underpriced stocks. My primary tool is the Price-to-Book (**P/B**) ratio. This ratio connects a company's stock price directly to its **Book Value of Equity**. The book value represents the net asset value on its balance sheet:

Book Value of Equity = Total Assets – Total Liabilities

Financial accounting relies on historical cost. Corporate assets are typically recorded at their original cost minus accumulated depreciation. This avoids speculative market estimates. Book value is a solid floor for what a company is worth on paper. It doesn't move with market hype.

I divide a company's total market capitalization by this concrete baseline to find the **P/B ratio**. A stock trading at a low P/B ratio near or below 1.0 is a "value stock." This idea is compelling. A low ratio suggests the stock is very cheap relative to its actual net assets. This offers a built-in "margin of safety."

But this traditional model has a major catch. A low P/B ratio can signal a **value trap**. A company might look cheap only because its business is failing or buried in debt. To avoid this, I add profitability metrics to the P/B filter: **Return on Assets (ROA)** and **Return on Equity (ROE)**. They show how well a company turns its assets and its shareholders' money into profit. This helps filter out failing businesses and find real value.

Research Objective and Hypothesis

I want to see if simple accounting data can beat market noise. First, I use the P/B ratio to find cheap stocks. Then, I add ROA and ROE filters to make sure these companies are healthy and profitable.

My hypothesis: combining value and quality filters will find stocks that the market mispriced. If I am right, my basket of stocks should regularly beat the benchmark.

Financial returns are messy and don't fit the clean assumptions of textbook significance tests, so I use two practical checks:

- **Average return (μ):** does my basket earn a higher average quarterly return than the benchmark? ($\mu_{\text{Strategy}} > \mu_{\text{Benchmark}}$)
- **Alpha (α):** after accounting for how much my basket simply moves with its benchmark (its beta), is there extra return left over? ($\alpha > 0$)

I also look at CAGR, volatility, Sharpe ratio and maximum drawdown (explained below), and at how often a strategy beats its benchmark across all the settings I tried.

Data Sources

Accounting and price data (SimFin)

I used the SimFin Python API to get quarterly balance sheets, quarterly income statements, daily stock prices and sector labels for US companies. From these I calculated market cap, book value, P/B, ROA and ROE. *The free API only gives the last 5 years of data, so my analysis is limited to that window.* The prices run from October 2020 to October 2025.

I use the adjusted close price, which accounts for stock splits and dividends.

Benchmark ETFs (yfinance)

I used the yfinance library to download prices for the benchmark ETFs from Yahoo Finance. I also used it as a second source to double-check SimFin's data.

When does the backtest start?

A company's numbers only count once they are public. Companies published their quarterly numbers a median of 39 days after the quarter ended, and 95% within 75 days. On December 31, 2020, only 5% of trading companies had published numbers in my data, because it starts in late 2020. By March 31, 2021, 74% had. So my first basket is picked on **March 31, 2021**, and the last return ends on **September 30, 2025**. That is 18 quarterly returns, about 4.5 years, limited by SimFin's free 5-year data.

Investment Universe and Benchmarks

I test the strategy in two ways.

Part A: sector by sector. I evaluated 11 sectors and matched each one with its own benchmark ETF, for a direct, fair comparison:

Technology: XLK; **Healthcare:** XLV; **Financial Services:** XLF; **Consumer Defensive:** XLP; **Consumer Cyclical:** XLY; **Industrials:** XLI; **Energy:** XLE; **Basic Materials:** XLB; **Real Estate:** XLRE; **Utilities:** XLU; **Business Services:** IYZ.

The free data doesn't include each ETF's actual holdings, so a sector's stocks are all trading companies that SimFin labels with that sector.

Part B: one portfolio across all sectors. Each quarter I take the 500 biggest trading US stocks by market cap, my stand-in for the S&P 500, and compare with **SPY**. The list is rebuilt every quarter using only what was known at the time.

Methodology and Backtest Architecture

I built a Python Colab notebook to test my hypothesis. It turns accounting data into a trading strategy.

How the Picker Works

Rebalancing is triggered by date at **every calendar quarter end** (March 31, June 30, Sept 30, Dec 31), not on individual earnings release days. On each quarter-end date, the selection follows four steps:

- **Find the stocks:** all stocks in the sector (or the top 500) that are trading on that quarter-end date, using each company's latest financial report published on or before that date.

- **Filter:** apply that stage's rules — basic checks (known data, positive equity, price $\geq$ \$1), size floors (market cap, book value), value cutoffs (P/B), and profitability cutoffs (ROA, ROE), measured either across the whole pool or within each sector.
- **Rank the survivors:** sort by cheapest P/B (raw or sector-normalized), or in Part A's control runs by biggest market cap or biggest book value.
- **Cap the basket:** buy the top K stocks, with equal weight, and hold them until the next quarter end.

If no stock passes, the money sits in cash for that quarter. If a stock I own stops trading (it gets delisted), I sell it at its last traded price.

Data Alignment

Matching accounting data with stock prices was not straightforward. Companies do not release financial data on the day a quarter ends. For example, a quarter might end on December 31, but the public sees the report weeks later. I had to prevent look-ahead bias ("looking into the future").

Initially I added a fixed three-month delay to all financial data. Later I changed it to quarter-end rebalance date - each company uses the latest financial report that SimFin says was published on or before that date. If the publish date is missing or looks wrong (only 0.1% of reports), I fall back to 90 days after the quarter ended. This is more accurate, and it still only uses data that was public at that quarter end.

Using CAGR to Compare Ideas

My code compares each basket with its benchmark using:

- **CAGR (Compound Annual Growth Rate):** the average yearly growth rate over the test.
- **Volatility:** how much the quarterly returns swing, scaled to a year.
- **Sharpe Ratio:** return per unit of risk, $(\text{CAGR} - 2\% \text{ risk-free rate}) \div \text{volatility}$.
- **Max Drawdown:** the largest drop from a peak to a low point.
- **Alpha (α):** the return left over after accounting for how much the basket moves with its benchmark (beta).

Every strategy and benchmark is compared by CAGR. A strategy "beats" its benchmark if its CAGR is higher over the same 18 quarters. The "gap" is the difference in CAGR, in percentage points a year. Returns are in %, not dollars, and every basket is equal-weighted. I checked my own formulas against the QuantStats library. Total return, volatility, a simple Sharpe and max drawdown all matched exactly.

Multi-Sector Automation ("The Smarts")

I automated the whole process. Instead of writing separate code for each sector, one loop runs everything:

None

```
For each Sector ETF in my target list:
  For each quarter in the backtest:
    1. Find the stocks trading on this date
    2. Filter and rank them to select the top K stocks
    3. Calculate the return and log the results
    4. Record the Ticker Delta (stocks added or removed)
```

The log records, every quarter, how many stocks were available, how many passed each filter, what was bought and sold, and each stock's return. It is a great debugging tool. I can see exactly why the script bought or sold a company.

Experimental Design

I set up my backtests as an experiment with adjustable settings. In Part A, I tested:

- **5 filter stages:** Pure P/B by itself, Clean P/B (size and price filters), and Clean P/B with profit checks (ROA, ROE, and ROA + ROE).
- **4 basket sizes (K):** 5, 10, 50 and 100 stocks. This shows the balance between buying only the top-ranked stocks and spreading out the risk (though as the filters tighten across the stages, fewer stocks pass, so the larger basket sizes $K = 50$ and $K = 100$ end up holding the exact same stocks in most sectors).
- **3 rankings:** cheapest P/B, compared with two control groups: the biggest companies by market cap, and the biggest by book value.

Across the 11 sectors, that is **132 runs per stage** (11 sectors $\times$ 4 basket sizes $\times$ 3 rankings), or 660 runs across all 5 stages.

Experiment goal: my goal is not to find a single perfect setting. I want to see the broader trends: does a bigger basket lower risk, does a profit filter protect the portfolio, and which filters actually help?

Cleaning the Data: The Biggest Thing I Learned

Initially, I learned that over 40% of the companies in the free dataset were missing their financial statements. Later, I learned that the data that *is* there has two very different kinds of traps — and they need different fixes:

1. **Real market extremes (like CODQL and "zombie" companies):** Sometimes the numbers in the file are real, but they break simple ratios or small equal-weight baskets. A tiny penny stock (**CODQL**) jumped from \$0.01 to \$0.95 in one quarter (+9,400%) and skewed an entire 5-stock sector basket, while 643 "zombie" companies had negative net income *and* negative equity at some point, making their ROE look positive (a negative divided by a negative). These are fixed by **strategy filters**: requiring positive equity, a \$1 price floor, and minimum company size.

2. **Corrupted vendor numbers (like Dolby and Eargo):** Sometimes the raw numbers themselves are wrong. In an early run before cleanup, a 1,000× units mix-up in SimFin's share counts made **Dolby** show up with a \$7.56 trillion market cap (bigger than Apple, when Dolby was really about \$7.5 billion) and a P/B of 3,210. In another case, Ampio's total assets were 700–800× too high, and a single bad daily price print for **Eargo** right before delisting created a fake +2,622% quarter. These had to be cleaned **before** running the backtest:
 - **Time-series baseline check:** A company's share count or total assets almost never jumps 50× in a single quarter and then snaps back (Dolby's share count read 95.9M, then 95,820M, then 95.7M). I flagged and removed values 50× or more away from a company's last 3 good reports — removing 592 of 47,537 share counts (1.2%) and a small number of asset, liability, equity and net-income outliers.
 - **Price-blip filter:** I removed daily prices 3× or more away from the median of the 11 trading days around them (633 of 6.16 million prices, or 0.01%). That caught single-day bad prints like Eargo without touching multi-week moves like CODQL.
 - **Second-source check (and knowing when to stop):** Against Yahoo Finance, 99.8% of 11,819 top-500 stock-quarters matched within 1%, and the median share-count ratio for the 100 largest stocks was 1.000. Once the largest companies (Microsoft, Apple, Nvidia, Alphabet, Amazon) all checked out at their real sizes, I stopped adding rules so I wouldn't throw away real data. All checks are in Appendix B of the notebook.

Part A: Sector by Sector (vs Sector ETFs)

Each stage fixes a problem I found in the one before. The tables show the cheapest-P/B ranking. Across all 11 sectors, the average sector ETF returned 9.62% a year (the median was 9.09%).

Stage 1: Pure P/B

I wanted to see what happens with a pure Price-to-Book strategy. I bought the cheapest stocks in each sector with no safety nets: no quality filters, no size limits and no price floor.

Sector (ETF)	ETF CAGR	K = 5	K = 10	K = 50	K = 100	Volatility K = 5
Technology (XLK)	19.14%	68.44%	53.56%	26.31%	22.57%	354.04%
Energy (XLE)	18.64%	-5.11%	11.33%	20.75%	21.28%	28.43%
Financial Services (XLF)	12.68%	-22.73%	11.76%	11.64%	13.54%	38.63%

Sector (ETF)	ETF CAGR	K = 5	K = 10	K = 50	K = 100	Volatility K = 5
Industrials (XLI)	12.22%	17.76%	32.74%	21.63%	13.87%	158.79%
Utilities (XLU)	10.42%	-1.74%	3.35%	5.84%	5.84%	22.82%
Consumer Cyclical (XLY)	9.09%	-71.56%	-49.82%	-17.63%	-10.38%	67.32%
Consumer Defensive (XLP)	5.84%	16.90%	3.35%	-2.48%	2.81%	162.54%
Healthcare (XLV)	5.66%	-59.71%	-54.30%	-7.07%	-7.69%	62.50%
Basic Materials (XLB)	4.98%	41.50%	61.94%	42.29%	29.71%	886.64%
Real Estate (XLRE)	4.83%	42.16%	35.54%	8.46%	6.91%	176.26%
Business Services (IYZ)	2.27%	-22.63%	-11.41%	-3.43%	-3.43%	29.63%

Bold = beat the sector ETF.

Impact of Basket Size

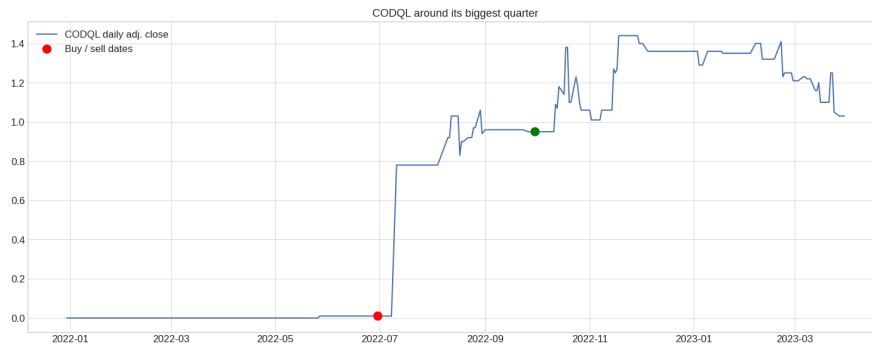
Basket size had a direct impact on risk. Averaged across the 11 sectors, the 5-stock cheapest-P/B baskets had a volatility of 180.69%. At 10 stocks it was 102.10%, at 50 stocks 33.14%, and at 100 stocks 24.55%. A larger basket protects the portfolio from single-company wild swings.

Impact of Ranking

- **The value trap:** ranking by cheap P/B alone led straight to distressed companies. Healthcare (-59.71%) and Consumer Cyclical (-71.56%) lost most of their money at K = 5.
- **Large-cap safety:** averaged across sectors, a 5-stock basket of the biggest companies made 11.62% a year with 20.82% volatility. The 5 cheapest stocks made 0.30% with 180.69% volatility. In my data, the biggest companies were much safer.
- **Winners:** pure P/B beat its ETF at every basket size in Technology, Industrials, Basic Materials and Real Estate. As the next section shows, much of that came from penny stocks.

The CODQL spike

The biggest single result came from one stock. In Basic Materials, a penny stock called **CODQL** jumped from \$0.01 to \$0.95 in one quarter (Q3 2022), a **9,400%** gain. That one stock made the 5-stock Basic Materials basket look like a 41.50% a year winner, with a crazy 886.64% volatility. Other big single-stock quarters were UAVS (+3,756%), QUBT (+2,446%) and RGTI (+1,856%).



CODQL's daily price. The red dot is when the basket bought it, the green dot is when it sold.

Other Interesting Findings

In small sectors like Utilities and Business Services, the 50-stock and 100-stock baskets had identical results. Those sectors only had about 27 to 29 stocks with usable numbers, so both baskets bought the same companies.

Across all 132 runs of this stage, pure P/B beat the sector ETF **35%** of the time. The median run trailed its ETF by 1.82% a year.

Stage 2: "Clean" P/B

Next, I got rid of tiny companies, weird outliers and businesses that barely own anything. Within each sector:

- **Market cap:** drop the smallest 25% of companies.
- **Book value:** drop the bottom 50% of companies by book value.
- **P/B cutoff:** only keep the cheaper half by P/B.
- **Price floor:** a \$1 minimum stock price, to avoid penny stocks.

Sector (ETF)	ETF CAGR	K = 5	K = 10	K = 50	K = 100	Stocks held K = 100
Technology (XLK)	19.14%	3.39%	-2.62%	5.66%	4.25%	85.9
Energy (XLE)	18.64%	15.74%	24.67%	18.85%	18.85%	22.6
Financial Services (XLF)	12.68%	17.46%	11.23%	6.73%	6.73%	31.0
Industrials (XLI)	12.22%	13.66%	14.93%	6.34%	7.03%	57.8

Sector (ETF)	ETF CAGR	K = 5	K = 10	K = 50	K = 100	Stocks held K = 100
Utilities (XLU)	10.42%	1.50%	2.07%	2.07%	2.07%	7.2
Consumer Cyclical (XLY)	9.09%	-6.15%	-4.51%	0.05%	-0.35%	57.9
Consumer Defensive (XLP)	5.84%	6.43%	1.00%	-1.06%	-1.06%	22.2
Healthcare (XLV)	5.66%	5.40%	4.61%	0.97%	-6.60%	99.6
Basic Materials (XLB)	4.98%	24.15%	17.33%	15.10%	15.10%	19.9
Real Estate (XLRE)	4.83%	-2.70%	-2.35%	-1.12%	-1.12%	27.1
Business Services (IYZ)	2.27%	-19.73%	-16.19%	-16.19%	-16.19%	7.4

What Changed

- **More realistic returns:** the 5-stock Basic Materials basket dropped from 41.50% to 24.15% a year, and its volatility from 886.64% to 30.02%. Its best stock was now VHI (+88%) instead of CODQL (+9,400%). The huge earlier return was an illusion caused by one penny stock.
- **The crashes got smaller:** Healthcare at K = 5 went from -59.71% to +5.40%, and Consumer Cyclical from -71.56% to -6.15%. Getting rid of tiny, distressed companies removed the worst value traps.
- **But the wins got smaller too:** Technology went from the best sector to beating its ETF in none of the 4 basket sizes. Its big pure-P/B returns had also come from tiny stocks.

Over-Filtering Starts

After these filters, only about 7 stocks were left in Utilities and Business Services. In most sectors, the 50- and 100-stock baskets bought the same stocks. This is the "over-filtering" problem I noticed initially.

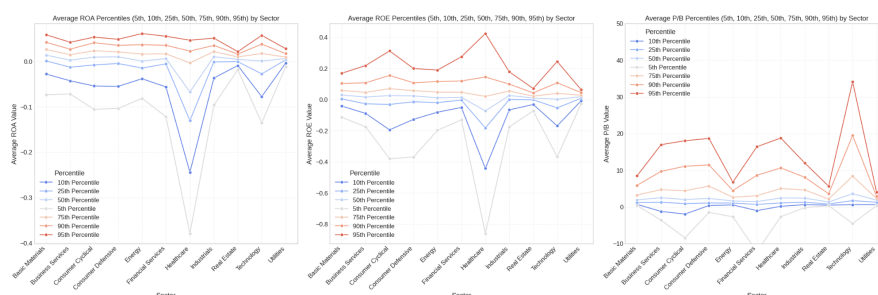
Overall, clean P/B beat the sector ETF only **23%** of the time (median gap -5.96% a year). Removing the penny stocks made the strategy safer, but it also removed the random-chance spikes that made pure P/B look good.

Stage 3: P/B + ROA

Clean P/B still didn't check whether a company actually makes money, so value traps could still get in. So I added one rule to the clean filters. Every stock must have a Return on Assets of at least 0.1% for the quarter before I rank it by value:

$$\text{ROA} = \text{Net Income} \div \text{Total Assets}$$

I used an absolute floor for every sector instead of a sector percentile. The chart below shows why. In Healthcare, the median ROA is below zero, so a percentile cutoff would let in money-losing companies. In Utilities, the values bunch up near zero, so a percentile cutoff would throw out healthy companies.



Sector (ETF)	ETF CAGR	K = 5	K = 10	K = 50	K = 100	Volatility K = 5
Technology (XLK)	19.14%	1.57%	-2.78%	7.18%	6.75%	27.08%
Energy (XLE)	18.64%	21.33%	14.10%	12.67%	12.67%	25.85%
Financial Services (XLF)	12.68%	4.87%	8.70%	5.91%	5.91%	19.60%
Industrials (XLI)	12.22%	8.89%	10.31%	8.65%	8.65%	19.85%
Utilities (XLU)	10.42%	4.78%	4.54%	4.54%	4.54%	14.44%
Consumer Cyclical (XLY)	9.09%	-5.27%	-2.08%	3.73%	3.52%	28.85%
Consumer Defensive (XLP)	5.84%	-1.38%	-0.03%	-0.85%	-0.85%	19.13%
Healthcare (XLV)	5.66%	-14.70%	-14.43%	-7.25%	-7.25%	35.06%
Basic Materials (XLB)	4.98%	26.19%	14.82%	17.39%	17.39%	23.29%
Real Estate (XLRE)	4.83%	-3.88%	-2.83%	-1.10%	-1.10%	23.00%
Business Services (IYZ)	2.27%	-21.55%	-20.12%	-20.12%	-20.12%	21.19%

What Changed Relative to Earlier Runs

- **Lower risk:** averaged across sectors, the 5-stock basket's volatility fell from 32.35% (clean P/B) to 23.39%. The ROA filter seems to have removed some shaky companies.
- **But not higher returns:** the average 5-stock basket made 1.89% a year, compared with 5.38% for clean P/B and 9.62% for the ETFs. The win rate fell to **13%**.
- **Healthcare is still broken:** at K = 5 it lost 14.70% a year. Even profitable cheap healthcare companies did badly in this period.
- **Winners:** Basic Materials beat its ETF at every basket size (26.19% at K = 5 vs. 4.98%), and Energy at K = 5 (21.33% vs. 18.64%).
- **Even more over-filtering:** with the profit rule, Utilities averaged only 6 stocks and Business Services fewer than 5. The 50- and 100-stock baskets had the same return in 9 of 11 sectors.

This is the result that changed most from my initial runs. Initially, adding ROA looked like a huge win: the 5-stock value portfolio made 37.79% a year across all sectors. Later, after tightening the data and making sure every filter really applies (see Challenges), the ROA filter mostly lowered risk instead of raising returns.

Stage 4: P/B + ROE

Now I look at Return on Equity (**ROE**). This metric tracks how well a company creates profits from the money shareholders invest.

ROE = Net Income ÷ Shareholders' Equity

ROE is very popular, but it has a flaw. A company can take on a lot of debt to shrink its equity. This pumps up ROE without any real business improvement. I call this the "leverage trap". In this test, I swapped the ROA rule for an ROE rule of at least 0.1%.

Sector (K = 5)	ROA CAGR	ROE CAGR	Difference (ROE – ROA)	ROE Volatility	ROE Sharpe
Financial Services	4.87%	-4.52%	-9.39%	21.68%	-0.30
Consumer Cyclical	-5.27%	-9.31%	-4.03%	30.43%	-0.37
Technology	1.57%	-0.72%	-2.29%	26.91%	-0.10
Consumer Defensive	-1.38%	-3.39%	-2.01%	19.99%	-0.27
Real Estate	-3.88%	-4.61%	-0.73%	22.91%	-0.29
Utilities	4.78%	5.17%	+0.39%	14.41%	0.22

Sector (K = 5)	ROA CAGR	ROE CAGR	Difference (ROE – ROA)	ROE Volatility	ROE Sharpe
Basic Materials	26.19%	26.76%	+0.57%	23.32%	1.06
Energy	21.33%	22.92%	+1.59%	25.12%	0.83
Business Services	-21.55%	-19.39%	+2.16%	21.00%	-1.02
Industrials	8.89%	11.51%	+2.62%	22.70%	0.42
Healthcare	-14.70%	-10.32%	+4.38%	31.90%	-0.39

Bold = ROE did worse than ROA.

Key Observations

- **This time it was close to a tie.** ROE did worse than ROA in 5 sectors and better in 6. Across all 132 runs, ROE beat the ETF 12% of the time and ROA 13%. Initially, ROE was worse in every sector I compared.
- **The biggest gap was in Financial Services** (-9.39% a year). Initially, I thought banks were the one sector where ROE works fine, because they use debt as a normal business tool. Later, the opposite happened.
- **Why the leverage trap is weaker now:** my picker now requires positive equity. That blocks the worst ROE glitch I saw initially, where "zombie" companies with negative income and negative equity got a positive ROE (a negative divided by a negative). Without those, ROA and ROE mostly pick similar companies.

Conclusion

ROE was not clearly worse than ROA this time. I still prefer ROA as the main profit check, because it can't be pumped up by debt and it doesn't break when equity is negative.

Stage 5: Stacking the Filters (P/B + ROA + ROE)

Finally, I required both ROA and ROE of at least 0.1%. The result was **exactly the same** as ROA alone, in every sector and basket size. When I checked, the reason was simple math. With positive equity, equity is always smaller than total assets. So ROE (net income ÷ equity) is always at least as big as ROA (net income ÷ assets). If ROA passes 0.1%, ROE does too, so the extra check removes nothing.

Project Progression Summary

Stage	Beat sector ETF (CAGR)	Median CAGR gap vs. ETF	K = 10 median CAGR	K = 10 median volatility (ETF: 17.23%)	K = 10 median max drawdown (ETF: -21.12%)
1. Pure P/B	35%	-1.82%	11.33%	77.02%	-41.86%
2. Clean P/B	23%	-5.96%	2.07%	24.84%	-38.06%
3. Clean + ROA	13%	-5.98%	-0.03%	19.76%	-29.32%
4. Clean + ROE	12%	-5.59%	-1.43%	20.14%	-34.28%
5. Clean + ROA + ROE	13%	-5.98%	-0.03%	19.76%	-29.32%

The median sector ETF made 9.09% a year (average 9.62%), with 17.23% volatility and a -21.12% max drawdown.

- **Pure P/B:** sorting by cheapness alone led straight to penny stocks and failing businesses. Its best results came from one-off spikes in tiny penny stocks (like CODQL's single-quarter +9,400% jump), where a single trade skewed the whole multi-year CAGR by random chance.
- **Clean P/B:** size filters and a price floor removed the penny stocks and cut the wild swings (median K = 10 volatility fell from 77.02% to 24.84%), but also removed the wins that came from random chance.
- **Clean P/B + ROA:** the profit check calmed the baskets further (median K = 10 volatility 19.76%, median max drawdown -29.32%; across all 132 runs in Stage 3, 46% had lower volatility than their sector ETF and 27% had a smaller max drawdown), but it did not raise returns, and the smaller sectors ran out of stocks.
- **Some settings outperformed, but confidence was low:** in every stage, some parameter settings beat their sector index (for example in Basic Materials and Energy), but with only 18 quarterly samples out of 132 runs per stage, confidence in any single setting was too low to trust it.

Which Variables Mattered Across the 660 Runs? (Logistic Regression)

Across all 5 stages of Part A, I ran 660 backtests (11 sectors × 5 stages × 4 basket sizes × 3 rankings), and 127 of them (19.2%) beat their sector ETF. To see which of my four choices actually mattered, I fit a simple logistic regression to predict whether a run beat its sector ETF (1 = beat ETF, 0 = trailed) and checked the R² for all four variables together and for each variable on its own:

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Variable	R ²
All 4 variables together	54.1%
Sector	42.9%
Filter stage	5.1%
Basket size (K)	2.0%
Ranking metric	0.2%

- **Which sector I was in did almost all the work ($R^2 = 42.9\%$).** Basic Materials beat XLB in 95% of its runs (57 of 60), followed by Industrials (30%) and Energy (30%). Those three asset-heavy sectors — where book value still reflects physical plants, mines and equipment — accounted for **73% of all wins** across the 660 runs (93 of 127), while Healthcare (0%), Utilities (1.7%) and Business Services (1.7%) almost never beat their ETF.
- **Filter stage ($R^2 = 5.1\%$) and basket size ($R^2 = 2.0\%$) had small effects.** Tightening the filters lowered the win rate (from 34.8% in Pure P/B to 12.9% in Clean + ROA) because removing penny stocks removed the one-off lottery spikes, and 5-stock baskets beat the ETF more often than 50-stock baskets (28.5% vs. 13.9%) because a single jumping stock can carry a tiny basket.
- **Ranking by cheapest P/B added almost nothing ($R^2 = 0.2\%$).** Across all 660 runs, ranking by cheapest P/B won 20.9% of the time (46/220), almost the same as picking the biggest companies by market cap (20.0%, 44/220) or by book value (16.8%, 37/220).

Part B: Running Against the Broad Market (SPY)

When I looked over my sector experiments, I noticed two clear patterns:

- **Small pools:** in small sectors like Utilities, there were not enough companies. After the quality filters, the pool dried up.
- **Useless ranking:** when the pool dried up, my P/B ranking stopped working. If only 6 stocks pass, a 50-stock basket buys all 6. The ranking doesn't matter at all.

I realized I needed a larger pool of companies. So I tested the 500 biggest US stocks together, against SPY.

Building it the other way around

Initially, I stacked all my filters at once and looked at the result. Later, I built a **trade-ready table**: one row per stock per quarter, with its P/B, ROA, ROE, market cap, book value, price, sector and what it returned over the next quarter. Then I added one filter at a time, a "filter ladder", and watched what each step did. The goal was to **weed out obvious losers**, not to guess the winners. At the top of the ladder are my strictest initial broad-market filters.

Step	Adds	Why
1. Pure P/B (with price $\geq$ \$1)	positive equity, P/B and market cap known, price $\geq$ \$1	can't value it, or a penny stock
2. + P/B $\leq$ 50th pct	P/B $\leq$ 50th percentile	it's a value strategy
3. + ROA $\geq$ 0.1%	ROA $\geq$ 0.1%	weed out money-losers
4. + ROE $\geq$ 0.1%	ROE $\geq$ 0.1%	same as Part A
5. + Size filters (= Part A Stage 3/5)	market cap $\geq$ 25th pct, book value $\geq$ 50th pct	Part A's "clean" filters
6. + ROA & ROE $\geq$ 75th pct	ROA and ROE $\geq$ 75th percentile (top 25%)	now hunting for winners
7. + P/B $\leq$ 25th pct (= initial strict)	P/B $\leq$ 25th percentile (cheapest 25%)	= my initial strict filters

Before trusting the table, I ran 8 settings through both the table and my Part A backtest engine. Every quarterly return matched exactly.

Shifting to Sector-Relative Percentiles

Financial numbers look very different from one sector to another:

- **Technology**: tech companies usually have high ROA, ROE and P/B ratios, because they don't need huge physical investments to grow.
- **Utilities**: utilities are very asset-heavy. They need massive power grids and plants, which leads to naturally lower ROA, ROE and P/B.

If I use the same cutoff for everything, the filters will favor sectors that are naturally cheap on paper. Initially, I fixed this with a peer-ranking strategy I called **Sector Neutrality**: each stock is ranked only against its own sector. Later, I ran every step both ways, to see if it actually helps:

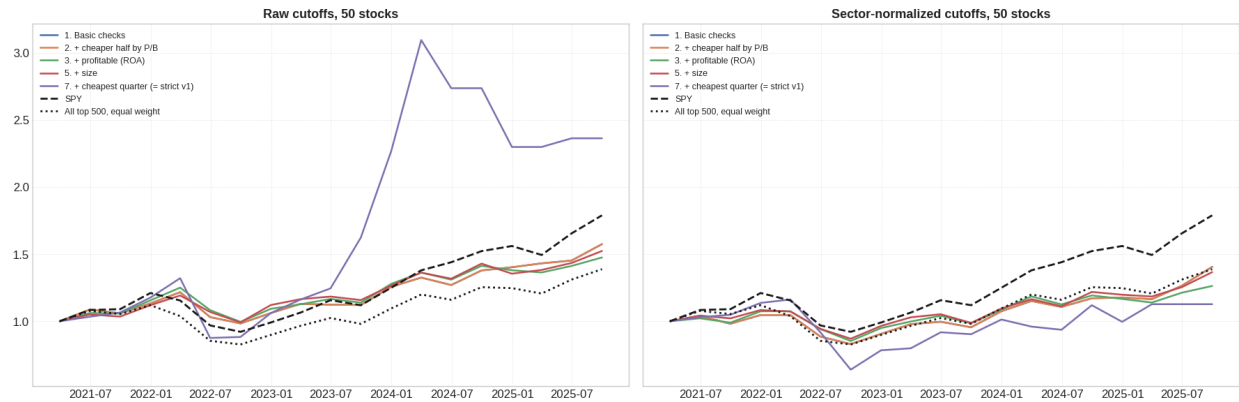
- **Raw**: percentiles measured against all 500 stocks. One cutoff for everyone.
- **Sector-normalized (Sector Neutrality)**: percentiles measured against the other top-500 stocks in the same sector, and baskets buy the cheapest stocks *within their sector*.

Broad Market Backtest Results

First, the two benchmarks. SPY made **13.81%** a year (volatility 15.11%, Sharpe 0.78, max drawdown -23.93%). Simply holding all 500 stocks with equal weight made only **7.55%**. That surprised me. In this period, a few giant companies (like Apple, Microsoft and Nvidia) drove the market. SPY holds them by size, while an equal-weight portfolio holds them the same as everyone else. So any equal-weight basket had to make up about 6 points a year just to match SPY.

Step	Stocks passing	50 stocks CAGR (raw / norm)	50 stocks Vol (raw / norm)	50 stocks Max DD (raw / norm)	All passing CAGR (raw / norm)	10 stocks CAGR (raw / norm)
SPY benchmark	500	13.81%	15.11%	-23.93%	13.81%	13.81%
All 500 (equal weight)	500	7.55%	15.41%	-26.16%	7.55%	7.55%
1. Pure P/B (with price $\geq$ \$1)	469	10.62% / 7.86%	13.30% / 14.62%	-19.15% / -20.75%	7.32% / 7.32%	3.14% / 5.06%
2. + P/B $\leq$ 50th pct	219	10.62% / 7.86%	13.30% / 14.62%	-19.15% / -20.75%	7.66% / 7.04%	3.14% / 5.06%
3. + ROA $\geq$ 0.1%	187	9.02% / 5.32%	13.43% / 13.81%	-20.43% / -20.66%	7.88% / 7.59%	-3.28% / 3.19%
4. + ROE $\geq$ 0.1%	187	9.02% / 5.32%	13.43% / 13.81%	-20.43% / -20.66%	7.88% / 7.59%	-3.28% / 3.19%
5. + Size filters (= Part A Stage 3/5)	122	9.81% / 7.17%	12.68% / 13.67%	-16.79% / -19.75%	8.51% / 7.97%	2.40% / 3.95%
6. + ROA & ROE $\geq$ 75th pct	7	14.20% / 25.93%	20.49% / 20.78%	-21.65% / -22.19%	14.20% / 25.93%	15.75% / 25.25%
7. + P/B $\leq$ 25th pct (= initial strict)	2	21.07% / 2.69%	35.87% / 26.62%	-33.75% / -45.04%	21.07% / 2.69%	21.07% / 2.69%

CAGR and volatility per year. Each strategy cell shows Raw / Sector-normalized. Step 4 changes nothing, for the same reason as Part A Stage 5.



Growth of \$1 for steps 1, 2, 3, 5 and 7 (50-stock baskets). Step 6 is in the table above.

What I found

- **The weeding-out steps (1 to 5) reduced downside and volatility below SPY, at the cost of lower returns.** They never beat SPY on CAGR (landing between -3.28% and 10.62% a year), but for the 50-stock baskets every one of Steps 1 to 5 had lower volatility (12.68% to 14.62% vs. SPY's 15.11%) and a smaller max drawdown (-16.79% to -20.75% vs. SPY's -23.93% and equal-weight's -26.16%). Step 5 (+ Size filters) cut the worst drop the most, to -16.79% raw and -19.75% sector-normalized.
- **Only the two "hunting for winners" steps beat SPY on return.** The best was Step 6 (+ ROA & ROE $\geq$ 75th pct), sector-normalized: 25.93% a year with a Sharpe of 1.15 (vs. SPY's 0.78) and a slightly smaller max drawdown (-22.19% vs. -23.93%). Its average quarterly return was 6.44% vs. SPY's 3.56%, and its alpha was +9.80%, so it passed both of my checks. But with about 7 stocks and only 18 quarters, that is weak evidence. The raw version only just beat SPY on return (14.20%), and on risk-adjusted return it lost (Sharpe 0.60).
- **But those baskets are tiny.** Step 6 left only about 7 stocks on average, and in some quarters just 1 or 2. So the "50-stock" basket was really a 7-stock basket.
- **Adding the cheapest-quarter P/B filter on top of quality (Step 7) drove downside risk way up.** Because only about 2 stocks passed, max drawdown worsened to -33.75% (raw, volatility 35.87%, Sharpe 0.53) and -45.04% (sector-normalized, where CAGR also dropped from 25.93% to 2.69%).

What happened to my initial result?

Initially, my strict broad-market filters made **46.02%** a year with 50 stocks, which I called "a strange surprise". Later, the same filters left about **2 stocks** on average, and in some quarters **none**. With so few stocks, the result depends completely on random chance: 21.07% raw, but 2.69% sector-normalized. The initial number came from a different 5-year window and a looser setup (see Challenges). The real lesson is the one I noticed initially under "Over-Filtering", just much bigger than I thought. My strict filters didn't leave enough stocks to build a real portfolio.

Does each filter actually weed out losers?

For each step, I split the stocks that passed the step before into the ones this step **removed** and the ones it **kept**. Then I compared what they did over the next quarter. A good weeding filter removes stocks that go on to do worse.

Filter step	Kept minus removed, raw	Kept minus removed, normalized	Quarters kept beat removed (normalized)
1. Pure P/B (with price $\geq$ \$1)	-0.81%	-0.81%	39%
2. + P/B $\leq$ 50th pct	+0.04%	-0.18%	50%
3. + ROA $\geq$ 0.1%	+0.63%	+1.00%	44%
5. + Size filters (= Part A Stage 3/5)	+0.40%	+0.17%	67%
6. + ROA & ROE $\geq$ 75th pct	+1.80%	+4.46%	89%
7. + P/B $\leq$ 25th pct (= initial strict)	+4.74%	-5.59%	31%

Average difference in next-quarter return.

- **Keeping the cheaper half by P/B did about nothing.** The stocks it kept and the ones it threw out did almost the same.
- **Just being profitable helped a little** (+0.63% a quarter raw, +1.00% normalized), but not every quarter.
- **Being among the most profitable helped the most.** In the sector-normalized version, the top-quarter-quality stocks beat the ones removed by 4.46% a quarter, in 16 of 18 quarters (89%).
- **The basic checks removed stocks that did slightly better.** They mostly remove stocks with negative equity. My guess is that some of those companies bounced back in this period.

Raw vs. sector-normalized: what did they buy?

My guess was that raw cutoffs would pile into naturally cheap sectors and skip Technology. That was partly right. At step 1, the raw 10-stock basket was 7% Technology on average, compared with 30% for the sector-normalized basket. The raw basket held more Financials (14% vs. 0%), Utilities (8% vs. 0%) and Business Services (8% vs. 0%). At Step 7 (the strictest step), the raw basket was mostly Basic Materials (35%), Consumer Cyclical (30%) and Energy (30%).

But sector neutrality didn't always win. Averaged over my grid search, raw cutoffs made 12.84% and sector-normalized 9.72% (50-stock baskets). Sector neutrality only clearly helped when combined with the top-quarter quality filter.

The Main Finding: Profitability Beat Cheapness

Because each setting is just a filter on the trade-ready table, I could try every combination: 4 P/B cutoffs × 4 profit filters × size filters on/off × raw vs. sector-normalized × 3 basket sizes (all passing stocks, 50 and 10). That is 192 portfolios. Only 22% of them beat SPY.

Then I looked at the average CAGR of the 50-stock baskets for each P/B × ROA combination (averaged over raw vs. sector-normalized cutoffs and size filters on/off):

P/B filter / Profit filter	none	ROA ≥ 0.1%	ROA ≥ 50th pct	ROA ≥ 75th pct	Row average
none	9.15%	7.83%	11.07%	14.22%	10.57%
P/B ≤ 75th pct	9.15%	7.83%	11.07%	14.16%	10.55%
P/B ≤ 50th pct	9.15%	7.83%	10.95%	15.96%	10.97%
P/B ≤ 25th pct	9.27%	7.73%	14.88%	20.25%	13.03%
Column average	9.18%	7.81%	11.99%	16.15%	11.28%

Bold = higher than SPY (13.81% a year).

Moving across the columns (from ROA ≥ 0.1% to ROA ≥ 75th pct) moved the average result by about **8 points a year** (7.81% to 16.15%). Moving down the P/B rows moved it by about **2.5** (10.55% to 13.03%), and almost all of that came in the right-hand columns where the quality filter had already narrowed the pool. The size filters barely mattered (11.36% off vs. 11.20% on). The weed-out test tells the same story: the cheaper-half P/B step did nothing, and the top-quarter quality step did the most. In 2021–2025, "is this a good business?" told me more about future returns than "is this stock cheap?". Just being profitable (ROA ≥ 0.1%) helped only a little; being among the *most* profitable helped a lot. This fits the value-trap idea from my introduction: cheap companies are often cheap for a reason.

The best single setting by Sharpe (raw, P/B ≤ 25th percentile, ROA ≥ 75th percentile, size filters on) made 29.91% a year. But it held only about 5 stocks, and I picked it out of 192 tries. With only 18 quarters, I treat the grid as a way to see patterns, not to pick a winner.

Data Anomalies, Limitations, and Caveats

My backtest results are just simulations. Real data is messy and trading costs money. Here are the problems and limits I know about:

1. Missing and wrong data

Many companies are missing their financial statements entirely:

Data	Available companies	Missing companies
Sector mapping	6,278 (95.80%)	275 (4.20%)
Price data	5,888 (89.85%)	665 (10.15%)
Income statements	3,742 (57.10%)	2,811 (42.90%)
Balance sheets	3,742 (57.10%)	2,811 (42.90%)

Since my strategy needs P/B, ROA and ROE, a company with missing data can never be picked. Of the missing companies that had prices, 8% were penny stocks, so dropping them also cleaned up the pool a little. Companies with missing data may also be closer to trouble, so ignoring them could make my strategy look safer than it is ("selection bias"). On top of that, some of the numbers that *are* there were wrong (see "Cleaning the Data").

2. A short test period

I only have 18 quarters (about 4.5 years), from March 2021 to September 2025, because of SimFin's free data limits. This period was unusual: it started in the COVID recovery, saw interest rates spike in 2022, and ended in a tech and AI boom led by a few giant companies. That boom is one reason equal-weight portfolios struggled against SPY. I cannot prove any of these results would last.

3. Small baskets

The only strategies that beat SPY held about 7 stocks or fewer, sometimes just 1 or 2. One or two stocks jumping by random chance can make or break those results.

4. Hidden trading costs

I assumed trading is free and instant, at the closing price. Real trading has fees, taxes and the gap between buy and sell prices (the spread). Rebalancing 50 or 100 stocks every quarter would add real costs.

5. Variables I didn't test

- **Other value metrics:** I only used P/B. I don't know if P/E or EV/EBITDA would pick better stocks.
 - **Other quality metrics:** I only checked ROA and ROE. I ignored debt levels, and ROA and ROE use a single quarter of profit, which can be noisy. A trailing 12-month version would be smoother.
 - **Rebalancing timing:** I rebalanced every quarter. Monthly or yearly might give different results.
 - **Top 500 is a stand-in:** my top 500 by market cap is close to, but not the same as, the real S&P 500.
-

Conclusion

My hypothesis was that combining value and quality filters would find mispriced stocks that beat the benchmark. Here is how it held up:

1. **Cheap stocks alone (P/B): no.** Pure P/B beat its sector ETF 35% of the time, and its best results came from one-off penny-stock spikes like CODQL. Once those were removed, the win rate fell to 23%.
2. **Cheap + profitable (ROA $\geq$ 0.1%): no.** The profit and size checks lowered volatility and cut max drawdown below SPY (down to -16.79% vs. SPY's -23.93%), but they beat the sector ETF only 13% of the time and never beat SPY on return.
3. **Cheap + very profitable (top-quarter ROA and ROE): partly.** This is the best result I found. Sector-normalized, it made 25.93% a year vs. SPY's 13.81%, with a better Sharpe and positive alpha. But it held only about 7 stocks. My strictest initial filters (Step 7, even cheaper) also beat SPY in the raw version, with only about 2 stocks and a lot more risk.

What I believed at the start was that cheapness plus a safety net would work. What I found is that **the quality part did most of the work, and the cheapness part added little**. And equal-weighting all 500 big stocks lost to SPY by about 6 points a year, which is one more point for just buying the index.

Key Learnings and Takeaways

This project taught me that building an investment strategy is hard. It is not just about writing code or looking at high returns. Managing real-world data and understanding financial math takes time and thinking.

1. Data quality and cleaning

This was the biggest lesson, again. Initially, I learned that over 40% of the companies were missing their financial statements. Later, I learned that the numbers that are there have two kinds of traps: real penny-stock and negative-equity extremes, and corrupted vendor numbers

(like share counts off by 1,000 times). I learned to check data over time (a share count doesn't jump 1,000 times in one quarter), to confirm with a second source, and to remove bad numbers instead of guessing them. I also learned when to stop: every cleaning rule throws away some data. If you do not know the flaws in your data, you cannot trust your results.

2. Financial ratios can lie (the leverage trap)

I cannot take popular metrics like Return on Equity (ROE) at face value. In my first runs, my code picked "zombie" companies with negative net income and negative equity. A negative divided by a negative is a positive, so failing companies looked highly profitable. Now my picker requires positive equity, which blocks this glitch. I learned to check the raw numbers underneath the ratios.

3. Hidden anomalies can shape outcomes

A single stock can break an entire test. In my 5-stock portfolio, a penny stock named **CODQL** jumped from \$0.01 to \$0.95 in one quarter (+9,400%). It made my strategy look amazing, but it was a freak event. In an earlier run, a bad price print for Eargo did the same thing (+2,622%). Now I always look at the biggest single-stock returns before I trust a result.

4. Diversification is important

Bigger baskets were much calmer. In pure P/B, the average volatility fell from 180.69% with 5 stocks to 24.55% with 100 stocks. A larger portfolio also protects against data glitches: with 50 stocks, one freak stock jump gets drowned out. Later, though, the only strategies that beat SPY were small baskets. So bigger isn't always better. It's a trade-off.

5. Don't compare apples to oranges

Different sectors have very different balance sheets, so I used sector percentiles to make a fair comparison. Later, I tested whether this actually helps. It did change what the portfolio bought (a lot more Technology), and it helped a lot with the top-quarter quality filter (in one setting with about 7 stocks). But on average, raw cutoffs did better (12.84% vs. 9.72%). A good idea still needs to be tested.

6. Over-filtering

Initially, my filters were too strict. Later, I found out how strict: my initial broad-market filters left about 2 stocks on average, and sometimes none. A good strategy needs quality controls, but it also needs enough stocks to build a real portfolio. Building the filters one step at a time, and checking what each step removes, showed me which filters were worth it.

7. The ultimate trade-off: risk vs. return

There is no free lunch. Spreading your risk smooths out market swings, but it limits your top returns. Every choice in strategy design is a trade-off.

8. Brute-forcing parameters is impossible (my next step)

I cannot test every combination of settings by hand. Later, I ran a small grid search (192 portfolios), and it already showed the danger: with only 18 quarters, the "best" setting is probably random chance. I still plan to study **Bayesian optimization**, which searches settings more cleverly. But I learned that the bigger need is more data: a longer history, so that a good result has a chance to be real.

9. Synthesizing my interdisciplinary interests

My favorite part was bringing all my interests together. This project combined my accounting knowledge with my love for financial markets. It also used my skills in programming and data analysis. It was highly satisfying to see these different subjects form a working strategy.

Challenges

The fake giants

The 1,000× share-count bug (see "Cleaning the Data") only showed up because I printed the biggest companies in my top 500 and looked at them. A number that looks too good, or too big, is usually a bug. This one led to all of my data cleaning.

Filters that weren't really applied

Initially, the market cap, book value and P/B percentile filters weren't all being applied the way I thought. Later, I updated each filter to return *why* a stock was removed, and the notebook counts how many stocks each filter removes. On June 30, 2023, for example, the clean filters removed 454 stocks for a small market cap, 472 for a small book value and 717 for being too expensive.

Looking into the future

Initially, I used a fixed three-month delay for financial data. Later, on each quarter-end rebalance date, I used each company's latest report published on or before that date. This is more accurate, and it lets the backtest use new numbers sooner (a median of 39 days after the quarter ends).

Stocks that stopped trading

Initially, stocks that stopped trading could sit in the data with a frozen price. Later, I required a stock to have a price in the last 7 days to count as trading. If I own a stock that stops trading, I sell it at its last traded price.

Checking the shortcut

For Part B, I used a faster "trade-ready table" instead of my main backtest engine. Before trusting it, I ran 8 different settings through both, and every quarterly return matched. I also checked my metric formulas against the QuantStats library.